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Course Title : Computer and Digital Forensics

Lab Title: Lab 3 Data Carving with XXD, Binwalk and Scalpel

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Executive Summary

This investigation focused on data carving and file recovery from forensic disk images using industry-standard digital forensic tools. The objective was to demonstrate the recovery and examination of files when filesystem metadata may be deleted, unavailable, or unreliable.

The investigation involved hexadecimal analysis of a JPEG file using xxd, extraction and examination of metadata using ExifTool, filesystem and deleted-file analysis using The Sleuth Kit, embedded-content analysis using Binwalk, and file carving using Scalpel. Deleted files were identified and recovered from a FAT12 forensic image, while a FAT16 USB forensic image was examined at the partition, filesystem, and sector levels.

Scalpel was configured to identify JPEG files using their file signatures and successfully carved 17 JPEG files from the USB forensic image. The carved files were recorded in Scalpel's audit.txt, providing a repeatable record of the recovery process.

Cryptographic hashes, including SHA-256, were generated at appropriate stages to support evidence integrity and verification. The investigation demonstrated that useful file content can remain recoverable from forensic images even when normal filesystem metadata is missing or files have been deleted.

All analysis was performed on working copies or extracted evidence, while the original forensic evidence was preserved to maintain forensic integrity.

Lab Objective

The objectives of this laboratory exercise were to:

1. Understand the principles and importance of data carving and file recovery in digital forensics.
2. Examine binary data using hexadecimal analysis and identify common file signatures and 3. headers/footers.

Analyze JPEG files by identifying SOI (FF D8) and EOI (FF D9) markers.

4. Reconstruct a binary file from hexadecimal data and verify its integrity using cryptographic hashes.
5. Use The Sleuth Kit to examine forensic disk images, filesystem structures, deleted files, inodes, sectors, and recoverable data.
6. Use Binwalk to identify embedded data and file structures within a forensic image.
7. Configure and use Scalpel to carve deleted JPEG files directly from a raw USB forensic image.
8. Verify recovered evidence using MD5 and SHA-256 hashing where appropriate.
9. Document the forensic procedures, findings, screenshots, and limitations in a professional and repeatable manner.

Tools and Resources Used

The following tools and resources were used during the investigation:

Software and Forensic Tools

Kali Linux — forensic analysis operating environment.

The Sleuth Kit (TSK) — filesystem, partition, inode, sector, and deleted-file analysis.

Autopsy Forensic Browser — forensic case management and evidence examination.

Scalpel 1.60 — signature-based file carving and recovery.

Binwalk — identification of embedded file structures and signatures.

ExifTool — extraction and examination of JPEG metadata.

xxd — hexadecimal examination and binary file reconstruction.

7-Zip (7z) — extraction and examination of the compressed forensic evidence archive.

Linux hashing utilities (md5sum, sha256sum) — cryptographic integrity verification.

file utility — identification and validation of recovered file types.

strings — examination of readable strings within binary and recovered files.

Evidence and Resources

Ch01InChap01.dd — FAT12 forensic disk image used for filesystem and deleted-file analysis.

J_ub_law.jpg — JPEG sample used for hexadecimal, metadata, reconstruction, and Binwalk analysis.

120M.7z — compressed forensic evidence archive containing the USB image.

usb_fat_carving.001 — FAT16 USB forensic image used for Scalpel file carving.

/etc/scalpel/scalpel.conf — Scalpel configuration file containing the enabled JPEG carving signatures.

Evidence Integrity

Cryptographic hashing, particularly SHA-256, was used to document and verify the integrity of forensic evidence and recovered data throughout the investigation.

Methodology

The investigation followed a structured forensic workflow. Evidence integrity was established using cryptographic hashing before analysis. The JPEG sample was examined using hexadecimal analysis,

metadata extraction, file reconstruction, and Binwalk. The forensic disk images were analyzed using The Sleuth Kit to identify filesystem structures and deleted files. The USB image was examined and processed with Scalpel for signature-based JPEG carving. Recovered files were validated and documented using file identification, hashing, audit logs, and screenshots.

Screenshots and Evidence

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ wget -q https://www.dropbox.com/s/a0lhsmzeh68wk1i/NTFS.001

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ ls -la
total 256576
drwxrwxr-x 8 kali kali 4096 Sep 3 12:43 .
drwx----- 38 kali kali 4096 Sep 3 08:52 ..
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 evidence
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 hashes
-rw-rw-r-- 1 kali kali 262694912 Sep 3 12:44 NTFS.001
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 output
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 recovered
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 screenshots
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 working
```

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ ls -la
total 256552
drwxrwxr-x 2 kali kali 4096 Sep 3 13:32 .
drwxrwxr-x 8 kali kali 4096 Sep 3 13:32 ..
-rw-rw-r-- 1 kali kali 262694912 Sep 3 12:44 NTFS.001

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ md5sum NTFS.001
fa7eecd50a691ab3245653ae91b762b2 NTFS.001

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$
```

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ file NTFS.001
NTFS.001: DOS/MBR boot sector MS-MBR Windows 7 english at offset 0x163 "Invalid partition table" at offset 0x17b "Error loading operating system" at offset 0x19a "Missing operating system", disk signature 0x9940673b; partition 1 : ID=0x7, start-CHS (0x0,2,3), end-CHS (0x1e,254,63), startsector 128, 509952 sectors

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$
```

Part A

Download an image to kali

```

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensic_s/file_carving/usb_file/J_ub_law.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ ls -la
total 1688
drwxrwxr-x 8 kali kali 4096 Sep 3 16:29 .
drwx----- 38 kali kali 4096 Sep 3 08:52 ..
drwxrwxr-x 2 kali kali 4096 Sep 3 13:32 evidence
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 hashes
-rw-rw-r-- 1 kali kali 1692150 Sep 3 16:29 J_ub_law.jpg
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 output
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 recovered
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 screenshots
drwxrwxr-x 2 kali kali 4096 Sep 3 08:52 working

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ ls J_ub_law.jpg -l
-rw-rw-r-- 1 kali kali 1692150 Sep 3 16:29 J_ub_law.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ display J_ub_law.jpg

```

Record the original hashes

```

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ md5sum J_ub_law.jpg | tee hashes/J_ub_law_original.md5
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ sha256sum J_ub_law.jpg | tee hashes/J_ub_law_original.sha256
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ cat hashes/J_ub_law_original.md5
cat hashes/J_ub_law_original.sha256
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg

```

Original Evidence Hash Verification

Before analysis, the original J_ub_law.jpg file was verified using MD5 and SHA-256 cryptographic hashing. The recorded hashes establish a baseline for demonstrating the integrity of the evidence throughout the forensic examination.

File: J_ub_law.jpg

Size: 1,692,150 bytes

MD5: 83a360ac7f7e0ca318e5bfe39f95f137

SHA-256: 238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468

The hash values were saved in the hashes/ directory for later comparison with the reconstructed image.

Examine the JPEG header

```

(kali@kali)-[~/SBT-DF202_Lab3]
$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@..
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  ....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ....1...!.

(kali@kali)-[~/SBT-DF202_Lab3]
$ xxd J_ub_law.jpg | tail
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae  ..=R..0....."..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7  -.b.z...z.WPk.c.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981  .U.$...d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9  U.....c`A....0:.
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd  ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80  )O.J.Y..(.QZ.v:..
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45  ....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1  .*-$.....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8...o..Z.T.e
0019d1f0: d351 9ebf ffd9  ....Q....

```

Note: The xxd utility was used to examine the binary structure of J_ub_law.jpg. The beginning of the file contained the JPEG Start of Image (SOI) signature FF D8 at offset 0x00000000. This was immediately followed by FF E1, indicating an APP1 segment containing Exif metadata.

The end of the file contained the JPEG End of Image (EOI) signature FF D9, observed at the final bytes of the hexadecimal output. The presence of both the SOI and EOI markers confirms the expected JPEG file boundary structure.

These signatures are important in digital forensics because they can be used to identify and carve JPEG data even when normal file-system metadata is unavailable or damaged.

Create a plain hexadecimal dump

```

(kali@kali)-[~/SBT-DF202_Lab3]
$ xxd -p J_ub_law.jpg > working/J_ub_law.hex

(kali@kali)-[~/SBT-DF202_Lab3]
$ head working/J_ub_law.hex
ffd8fffe117f0457869666000049492a00080000001000000010300010000000
40130000001010300010000000d00c0000002010300030000000ce0000000601
0300010000000200000000f01020012000000d40000001001020009000000
e6000000120103000100000000100000015010300010000000030000001a01
050001000000ef0000001b01050001000000f70000002801030001000000
020000003101020021000000ff0000003201020014000000200100009882
02000c000000340100009e9c010052000000400100006987040001000000
94010000440400000800080008004e494b4f4e20434f52504f524154494f
4e004e494b4f4e204443400c0c62d0010270000c0c62d001027000041646f
62652050686f746f73686f702032312e3120284d6163696e746f73682900

(kali@kali)-[~/SBT-DF202_Lab3]
$ tail working/J_ub_law.hex
00fb4f29ff00194fcffc1d09766457dbae9fcd74d3f9ff0087aad8cb53bd
2bb488cde3b393a668a39ea35484ac487d4c7493e91fe361c5fdac15e271
d069f0dd324f45283e29c78ac8a18ca156232e93a63f347c298ed6be9b0e
4137fa68f4deaeb2d3f9ab54ac92c2f3685868a9669b4ad512eab3149471
c051235fd2c3817b7bf647cebd5abdbf67520186378d7cb306f29f5c28ea
ecfe3d5219183008da88b4897f22feae2def629c7ab516bc7acd57506ba9
63a791550524f1a8648f4d6af9c795e4898155d615198293636041e0fbd5
05303aa90bab554f4d91346a25a799d92a52a1bd294fae4aa859c9b42885
515af6763a80fc7f81df1c79f5e1dc0f53e3a4d55f45182ad52d2406abc1
ad6994790789d1e1ff0052540938040e6ff5f75a9a54f565d3519ebfffd9

```

A plain hexadecimal representation of J_ub_law.jpg was created using the xxd -p utility and saved as working/J_ub_law.hex. Examination of the dump showed the JPEG SOI signature FFD8 at the beginning of the file and the EOI signature FFD9 at the end.

The plain hexadecimal format removes the address offsets and ASCII display used by the standard xxd output, making it suitable for searching directly for hexadecimal file signatures and for subsequent reconstruction of the binary file.

JPEG Signature Search

```
(kali@kali)~/SBT-DF202_Lab3
$ grep -in "ffd8" working/J_ub_law.hex
1:ffd8ffe117f045786966000049492a000800000010000001030001000000
40:010000004c1300000000000048000000010000004800000001000000ffd8
285:005a000001e000000a8c00000134600180001ffd8ffed000c4164f62655f
4416:fdeffd8e9cfdc3bed17672ff00bcff00b3d23b7b749f70f5b63a9b37d89d
4438:88ff0037cffd8ebc7f675d1462490ba581b03a88bdd6f6361c0fa7bafcbce
8053:bcdffd87bf11434eb7aab9a75cb43ae9b00cca3594fcfa490508fa0fc922
6585:9edc7e9bb7f79b7111e8b0b1ce547d0f1cf1c83f8f79916ffd8454f244ff
6670:8ecea03cd21e743af4900dffd80f65db9d0da1271914fdbd29dbc7ead46
9252:11eafd20f1617b9d478b5ffd8fb870eb5d7480fd6c0e91a4f0a3a580b03c
9613:61700fd05ffd8ff43e0c08229d7abea3aecdcf3a594b2850a8c1bf6c90a6
10769:e829d68d3cfae17a481fd6c45cea7a81cf3cdffd87bb7954feced41d7
11046:815ad6a3ae045aea106ab9b13ea5047234a017b1fcdffd81e3defab10a3c
11066:7d42dffd89e3deb1c0756a1ceald75a748b36962406d3720480027d449e2
12044:2a8525f27a45ffd80b5c7f8fb12b0a069fcf2a1527f3e8b9fc9bfcba3a
15617:e0f3ab4a81f0f5f8f7eebc45475c34fd085d36620a92a38d1c5ffd87d7f2
15895:3ff0057e7d74da514b9d68096625b91736b5ffd80b13fd7fd6f7e35ad3a
15932:ae4927d5722c0ac66ee795209e41fc5ffd8ff53eea318eaf4a8c75c0fe94
19270:f7b7dd222a37cd95655a773432152a7ccac641d55f42d5a749affd80db0a
21911:040b5b81cfffad4f7ed63af63ac66fae6bea079b58dffd88febfeb7bb79f
22834:56ff004b5f8d8fffc53df88a007ab50f50ab081b8f6a5ee58e3b7a2865d40
24778:8722ff00d3deab4f55b5ebc57c7ffd8eb0937d23f511fd1afa6cdfd47d49e3
25213:c6b72410837252398f4fe7fadffd8f27dd4b3139f207fc23ad529435ea06
29643:c39b0b83c73cdffd81f7bf03e58ea9e2a70eb83641893600af20fd4a8046
31880:527957cb19543684a6e39f6ae7b208e559c3280294c57c7ffd8fb7a7ed1c
35324:bb014345f2fd9d34d83420ff009ffd8f6a44bc719a3b83af3cd9245e40d2b
42585:c7038e817dd9dc3b63776ffd89be697a43a8365e3b60c782153d518a8ed
44848:3a64cd7c24ee5df11d7657befb0fe346d9abd4d451b4362e7ffd8bd9f89a
52355:3e736b6ecce7a9dffd8d57f1637af537c839b6f65a933fd27b7e833f88cd

(kali@kali)~/SBT-DF202_Lab3
$ grep -in "ffd9" working/J_ub_law.hex
905:ff00e0ebf17f045786966000049492a00080000f746f73686f7020332e3000
450:ffd93842494d042100000000005700000001010000000f00410064006f00
1066:aeaf1feafcbaf11e9d712bc721745f862a48605d89b726c6ffd9fa7e79bf1
1863:a7fd0fadc71c8e7de0bd6eb83d762e18117d40093a001fd0f5158dc7d071
4061:927af6af5e3d7651a7ffd96e525ba87ea00adecbf937ff5fb03dfb1c7ad82
4283:f827ff05ff3efc07f9f5bd2c7ffd9fcbae84001b86b923553584c9
4485:fd037eaf7b0f4d60b0eb9a14baf8651e006e0a0e5d3f4235596ffd91
4637:105c803fa7e0ff004f7942a29281f31fe1ea0f6ae9cfa75b3af5c7ffd93de
4949:4933f85f3f85f3f857607800ff8baf7f00a37ff57ed8f6bcf37fd1c26
6503:795454d091f31e47d7a70529538ff275348ff376e6ffd9b1ff13fd9b5ed
8798:7d3cf3d21f620ffd93b877bed3d8dd61833fe8fb1749beb3837927dcd0e1
11024:55f053a887c7ffd90fb77f045786966000049492a00080000f746f73686f7020332e3000
11464:cbfcff02eb7d70603d2c7ffd960df6ebdc1e582f1feb81ef6092085ff0055
11759:77f4e920817fc8bdedc5fdeaa7ab6a6a54711fc7ffd9ebad05581f50be416
12788:7bd86ff579f8a0a2f6a3d308a393ad992ea2c7ffd9036f4b0e2f7f000faf
14848:c2e0858052c0325b5adcd10abf5fcfd381f9bfbdfcdffd9eb5f2eb90d4154a
16038:10a7dbfe0ead3f953ff00349f8brr0010rf23f2c700ff2320b6329ffd98fe
17485:00040e05e7ffd90b0f6bf64d097aa1f2eb8b9e48e0dc13f8d43f22c3fdff5f
17530:2d22dd87badc9014c8a698a7edf2a7c7ffd9e9c81bac8578d7cf1fcbf6797
19658:d5ff009774bad995622294ff0097c7ffd9e80a9f012a218853aee704fa1
22736:dafcc3fafb135460000c1cf4a7950c09c91c47407757f5feffd93d7bb471d
22958:af5c5fba01e7fe7eb8b07e792069040d60390ffd931fed42d724f3f8ff00d
33040:207e2f7b8e3fd87fadefc1a9c7af6927e5d71370bf5b8b0e39526ffd907c
25877:37d4d88bd88ff7d7f7e048e18ebda41a7cbac6ce41b82355ee6ffd9fea78
29348:ae1fd3ea936741729a6ffd903fa1fadfe83daf66d157952461826f66ff8
39659:ca9b6c7f00821e0a7f6d4937f6d4f31094f253ff670f3e9d828661c2a7a
31515:9b71d5e7ffd9055b8b7d76b6feaf5c74bf616c7db33769f616e9ec31b6364
32414:c0e3c7ffd9e8a0ad9f9dd45d386cc52f5cfcdddbcb47532622a017c47c2f
32686:f51fa230ff000d8a7a70e1d24ffd9e7f980e1d833af2c7f60ed032d36d2
37610:6c6c7ffd93f53ee3d4d904d01a75bed0c283fe2bfe2fa89344c444aeece9a0e
31866:24d1dc66b535f997f9fac741ef50e33be7ffd9380872b14990fe85365d
42677:1cca08553fcffd9ca239220aeb0c9f4a5ae7fc0308e90105dc787aa8c8
51653:ec7ffd9099cf6cb972cf0d155d4ed3f2e5696d14d134546c9e1c8d3527898a
1777:007f8790a7ffd934dad690a897a63b231b725ad14f529b532134a0482a2
52342:03a304791759147d219c0ab354d15d4004051d02d98962c76ffd91b4bbcc7
54345:05415f4dc7d432700588361ed14b2d246a8ae7fe2ffd9e9b76225d24803e
56405:ad6924790789d411ff0052540538040e6ffcf75a9a54f56cd3519ebfffd9
```

The plain hexadecimal dump was searched for the JPEG SOI (FFD8) and EOI (FFD9) signatures. Multiple occurrences of both byte sequences were identified within the dump. This demonstrates that signature patterns may occur inside compressed JPEG data and therefore cannot automatically be interpreted as file boundaries.

The valid JPEG boundaries were confirmed from the original xxd output: FFD8 occurs at offset 0x00000000 as the Start of Image marker, while FFD9 occurs at the end of the file as the End of Image marker.

Prepare the hex dump for reconstruction

```
(kali@kali)~/SBT-DF202_Lab3
$ tr -d '\n' < working/J_ub_law.hex > working/J_ub_law_singleline.hex

(kali@kali)~/SBT-DF202_Lab3
$ head -c 100 working/J_ub_law_singleline.hex
echo
ffd8ffe117f045786966000049492a000800000010000001030001000000401300000101030001000000d00c000002010300
```

Reconstruct the JPEG

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ xxd -r -p working/J_ub_law_singleline.hex working/J_ub_law_reconstructed.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ ls -lh working/J_ub_law_reconstructed.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  3 17:25 working/J_ub_law_reconstructed.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ file working/J_ub_law_reconstructed.jpg
working/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
```

Verify the hashes

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ md5sum J_ub_law.jpg working/J_ub_law_reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137  working/J_ub_law_reconstructed.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
└─$ sha256sum J_ub_law.jpg working/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  working/J_ub_law_reconstructed.jpg

(kali㉿kali)-[~/SBT-DF202_Lab3]
```

Hexadecimal Reconstruction and Integrity Verification

The JPEG file was converted to a hexadecimal representation and reconstructed using `xxd -r -p`. The reconstructed file was identified as valid JPEG image data with Exif metadata, including Nikon D4 camera information.

Integrity verification produced identical hashes for both files:

MD5: 83a360ac7f7e0ca318e5bfe39f95f137

SHA-256: 238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468

The matching MD5 and SHA-256 values confirm that the reconstructed JPEG is bit-for-bit identical to the original evidence file.

Part B


```

[ball@kali ~]$ cat /dev/urandom | tr -dc 'a-z0-9' | fold -w 64 | xargs sha1sum
ExifTool Version Number      12.05
File Name                    13.jpg
Directory                    3_000_000_000.jpg
Date/Time Original            1092 KB
File Modification Date/Time   2020:09:03 10:29:47-04:00
File Access Date/Time         2020:09:03 10:31:12-04:00
File Permissions              -rw-rw-r--
File Type                     JPEG
File Type Extension           jpg
MIME Type                    image/jpeg
Exif Byte Order               Little-endian (Intel, II)
XMP                            Not Comprehension
Camera Model Name             NIKON D4
Orientation                   Horizontal (normal)
Samples Per Pixel             3
Resolution                    300
X-Resolution                  140lines
Resolution Unit               inches
Modify Date                   2020:08:20 10:20:05
Keywords                      2013, campus, Merrick School of Business
XP Keywords                   2013, campus, Merrick School of Business
Exposure Time                 11.0
Exposure Program              Manual
Sensitivity Type              Recommended Exposure Index
Exif Version                  2.2
Date/Time Original            2013:10:08 18:50:21
Create Date                   2013:10:08 18:50:21
Shutter Speed Value           1/10
Aperture Value                2.8
Exposure Compensation         0
Max Aperture Value            2.8
Matrix Mode                   Unknown
Light Source                  Unknown
Flash                         On
Focal Length                  20.0 mm
Sub Sec Time Original         Not Calibrated
Exif Image Width              1920
Exif Image Height             1080
Focal Plane X Resolution      1368.888889
Focal Plane Y Resolution      1080
Focal Plane Resolution Unit   cm
Printing Scale                 cm
Scene Type                    on-chip color area
Exif Camera                   Directly Photographed
Color Space                    Normal
Custom Rendered               Normal
Exposure Mode                  Auto
White Balance                  10000
Digital Zoom Ratio            1.0
Scene Capture Type             Standard
Color Control                  Normal
Contrast                       Normal
Saturation                     Normal
Sharpness                     Unknown
Subject Distance Range        17-25mm f/2.8
Lens Info                      17-25mm f/2.8
Compression                   JPEG (old-style)
Thumbnail Offset              1100
Thumbnail Length              4934
Current IPRC Offset            7941212f0f8a22b8acc88b2b4633ef
Coded Character Set           UTF8
Configuration Record Version   00:00:00
Copyright Notice              2013, campus, Merrick School of Business
Copyright Owner               NIKON D4
Copyright User                 7941212f0f8a22b8acc88b2b4633ef
Displayable Units             inches
Displayable Units             Centimeters
Print Position                 0
Global Scale                   30
Global Angle                   30
Global Altitude               0
Silent Group Name             Low_1_room
Pixel Aspect Ratio            1
ExifTool Version Number      12.05
File Name                    13.jpg
Directory                    3_000_000_000.jpg
Date/Time Original            1092 KB
File Modification Date/Time   2020:09:03 10:29:47-04:00
File Access Date/Time         2020:09:03 10:31:12-04:00
File Permissions              -rw-rw-r--
File Type                     JPEG
File Type Extension           jpg
MIME Type                    image/jpeg
Exif Byte Order               Little-endian (Intel, II)
XMP                            Not Comprehension
Camera Model Name             NIKON D4
Orientation                   Horizontal (normal)
Samples Per Pixel             3
Resolution                    300
X-Resolution                  140lines
Resolution Unit               inches
Modify Date                   2020:08:20 10:20:05
Keywords                      2013, campus, Merrick School of Business
XP Keywords                   2013, campus, Merrick School of Business
Exposure Time                 11.0
Exposure Program              Manual
Sensitivity Type              Recommended Exposure Index
Exif Version                  2.2
Date/Time Original            2013:10:08 18:50:21
Create Date                   2013:10:08 18:50:21
Shutter Speed Value           1/10
Aperture Value                2.8
Exposure Compensation         0
Max Aperture Value            2.8
Matrix Mode                   Unknown
Light Source                  Unknown
Flash                         On
Focal Length                  20.0 mm
Sub Sec Time Original         Not Calibrated
Exif Image Width              1920
Exif Image Height             1080
Focal Plane X Resolution      1368.888889
Focal Plane Y Resolution      1080
Focal Plane Resolution Unit   cm
Printing Scale                 cm
Scene Type                    on-chip color area
Exif Camera                   Directly Photographed
Color Space                    Normal
Custom Rendered               Normal
Exposure Mode                  Auto
White Balance                  10000
Digital Zoom Ratio            1.0
Scene Capture Type             Standard
Color Control                  Normal
Contrast                       Normal
Saturation                     Normal
Sharpness                     Unknown
Subject Distance Range        17-25mm f/2.8
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Thumbnail Offset              1100
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Coded Character Set           UTF8
Configuration Record Version   00:00:00
Copyright Notice              2013, campus, Merrick School of Business
Copyright Owner               NIKON D4
Copyright User                 7941212f0f8a22b8acc88b2b4633ef
Displayable Units             inches
Displayable Units             Centimeters
Print Position                 0
Global Scale                   30
Global Angle                   30
Global Altitude               0
Silent Group Name             Low_1_room
Pixel Aspect Ratio            1
ExifTool Version Number      12.05
File Name                    13.jpg
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File Permissions              -rw-rw-r--
File Type                     JPEG
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Exif Byte Order               Little-endian (Intel, II)
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Camera Model Name             NIKON D4
Orientation                   Horizontal (normal)
Samples Per Pixel             3
Resolution                    300
X-Resolution                  140lines
Resolution Unit               inches
Modify Date                   2020:08:20 10:20:05
Keywords                      2013, campus, Merrick School of Business
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Exposure Program              Manual
Sensitivity Type              Recommended Exposure Index
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Shutter Speed Value           1/10
Aperture Value                2.8
Exposure Compensation         0
Max Aperture Value            2.8
Matrix Mode                   Unknown
Light Source                  Unknown
Flash                         On
Focal Length                  20.0 mm
Sub Sec Time Original         Not Calibrated
Exif Image Width              1920
Exif Image Height             1080
Focal Plane X Resolution      1368.888889
Focal Plane Y Resolution      1080
Focal Plane Resolution Unit   cm
Printing Scale                 cm
Scene Type                    on-chip color area
Exif Camera                   Directly Photographed
Color Space                    Normal
Custom Rendered               Normal
Exposure Mode                  Auto
White Balance                  10000
Digital Zoom Ratio            1.0
Scene Capture Type             Standard
Color Control                  Normal
Contrast                       Normal
Saturation                     Normal
Sharpness                     Unknown
Subject Distance Range        17-25mm f/2.8
Lens Info                      17-25mm f/2.8
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Copyright Notice              2013, campus, Merrick School of Business
Copyright Owner               NIKON D4
Copyright User                 7941212f0f8a22b8acc88b2b4633ef
Displayable Units             inches
Displayable Units             Centimeters
Print Position                 0
Global Scale                   30
Global Angle                   30
Global Altitude               0
Silent Group Name             Low_1_room
Pixel Aspect Ratio            1
ExifTool Version Number      12.05
File Name                    13.jpg
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Modify Date                   2020:08:20 10:20:05
Keywords                      2013, campus, Merrick School of Business
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Exposure Time                 11.0
Exposure Program              Manual
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Aperture Value                2.8
Exposure Compensation         0
Max Aperture Value            2.8
Matrix Mode                   Unknown
Light Source                  Unknown
Flash                         On
Focal Length                  20.0 mm
Sub Sec Time Original         Not Calibrated
Exif Image Width              1920
Exif Image Height             1080
Focal Plane X Resolution      1368.888889
Focal Plane Y Resolution      1080
Focal Plane Resolution Unit   cm
Printing Scale                 cm
Scene Type                    on-chip color area
Exif Camera                   Directly Photographed
Color Space                    Normal
Custom Rendered               Normal
Exposure Mode                  Auto
White Balance                  10000
Digital Zoom Ratio            1.0
Scene Capture Type             Standard
Color Control                  Normal
Contrast                       Normal
Saturation                     Normal
Sharpness                     Unknown
Subject Distance Range        17-25mm f/2.8
Lens Info                      17-25mm f/2.8
Compression                   JPEG (old-style)
Thumbnail Offset              1100
Thumbnail Length              4934
Current IPRC Offset            7941212f0f8a22b8acc88b2b4633ef
Coded Character Set           UTF8
Configuration Record Version   00:00:00
Copyright Notice              2013, campus, Merrick School of Business
Copyright Owner               NIKON D4
Copyright User                 7941212f0f8a22b8acc88b2b4633ef
Displayable Units             inches
Displayable Units             Centimeters
Print Position                 0
Global Scale                   30
Global Angle                   30
Global Altitude               0
Silent Group Name             Low_1_room
Pixel Aspect Ratio            1
ExifTool Version Number      12.05
File Name                    13.jpg
Directory                    3_000_000_000.jpg
Date/Time Original            1092 KB
File Modification Date/Time   2020:09:03 10:29:47-04:00
File Access Date/Time         2020:09:03 10:31:12-04:00
File Permissions              -rw-rw-r--
File Type                     JPEG
File Type Extension           jpg
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Exif Byte Order               Little-endian (Intel, II)
XMP                            Not Comprehension
Camera Model Name             NIKON D4
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Samples Per Pixel             3
Resolution                    300
X-Resolution                  140lines
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Exposure Compensation         0
Max Aperture Value            2.8
Matrix Mode                   Unknown
Light Source                  Unknown
Flash                         On
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Sub Sec Time Original         Not Calibrated
Exif Image Width              1920
Exif Image Height             1080
Focal Plane X Resolution      1368.888889
Focal Plane Y Resolution      1080
Focal Plane Resolution Unit   cm
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Custom Rendered               Normal
Exposure Mode                  Auto
White Balance                  10000
Digital Zoom Ratio            1.0
Scene Capture Type             Standard
Color Control                  Normal
Contrast                       Normal
Saturation                     Normal
Sharpness                     Unknown
Subject Distance Range        17-25mm f/2.8
Lens Info                      17-25mm f/2.8
Compression                   JPEG (old-style)
Thumbnail Offset              1100
Thumbnail Length              4934
Current IPRC Offset            7941212f0f8a22b8acc88b2b4633ef
Coded Character Set           UTF8
Configuration Record Version   00:00:00
Copyright Notice              2013, campus, Merrick School of Business
Copyright Owner               NIKON D4
Copyright User                 7941212f0f8a22b8acc88b2b4633ef
Displayable Units             inches
Displayable Units             Centimeters
Print Position                 0
Global Scale                   30
Global Angle                   30
Global Altitude               0
Silent Group Name             Low_1_room
Pixel Aspect Ratio            1
ExifTool Version Number      12.05
File Name                    13.jpg
Directory                    3_000_000_000.jpg
Date/Time Original            1092 KB

```

[illegible]

ExifTool was used to examine embedded JPEG metadata. The file was identified as a JPEG produced by a NIKON D4 camera. The metadata records an original capture date of 8 October 2013 at 18:56:21.80, with an exposure of 8 seconds, f/11 aperture, ISO 200, and a 20 mm focal length.

The metadata identifies HOWARD KORN as the copyright holder and contains the keywords “2013, campus, Merrick School of Business.” The image was processed using Adobe Photoshop 21.1 (Macintosh), with metadata showing subsequent editing and conversion activity through 2020.

The metadata also contains a camera serial number, lens information, raw file name, document identifiers, and editing history. These fields may provide useful investigative leads but should be interpreted cautiously because metadata can be modified during image processing.

Part C

Sleuth Kit Filesystem and Data-Unit Examination

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ ls -lh evidence/Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Sep  3 17:58 evidence/Ch01InChap01.dd

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ sha256sum evidence/Ch01InChap01.dd | tee hashes/Ch01InChap01.sha256
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 evidence/Ch01InChap01.dd

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ file evidence/Ch01InChap01.dd
evidence/Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ", 'k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤32 MB), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT
```

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ img_stat evidence/Ch01InChap01.dd | tee working/Ch01InChap01_img_stat.txt

IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ mmls evidence/Ch01InChap01.dd | tee working/Ch01InChap01_mmls.txt
```

Examine the FAT12 filesystem

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ fsstat evidence/Ch01InChap01.dd | tee working/Ch01InChap01_fsstat.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT12
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 3M
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879
METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848
FAT CONTENTS (in sectors)
-----
33-236 (204) -> EOF
285-311 (27) -> EOF
```

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ fls -r evidence/Ch01InChap01.dd | tee working/Ch01InChap01_fls.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

Examine the deleted file metadata

From your fls output, the deleted files are:

- 8 — Billing Letter.doc
- 11 — confirmation.txt
- 15 — letter1.txt
- 17 — Regrets.doc

Let's examine inode 8 first.

```

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 8 | tee working/istat_8_Billing_Letter.txt
Directory Entry: 8
Not Allocated
File Attributes: File, Archive
Size: 24064
Name: _ILLIN-1.DOC

Directory Entry Times:
Written:      2005-12-09 06:50:28 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:05 (EST)

Sectors:
237 238 239 240 241 242 243 244
245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276
277 278 279 280 281 282 283

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 11 | tee working/istat_11_confirmation.txt
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR-1.TXT

Directory Entry Times:
Written:      2005-12-09 06:52:58 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:06 (EST)

Sectors:
284

```

```

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 15 | tee working/istat_15_letter1.txt
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:09 (EST)

Sectors:
312

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ istat evidence/Ch01InChap01.dd 17 | tee working/istat_17_Regrets.txt
Directory Entry: 17
Not Allocated
File Attributes: File, Archive
Size: 23552
Name: _EGRETS.DOC

Directory Entry Times:
Written:      2005-12-09 06:50:52 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:09 (EST)

Sectors:
313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328
329 330 331 332 333 334 335 336
337 338 339 340 341 342 343 344
345 346 347 348 349 350 351 352
353 354 355 356 357 358

```

Recover the deleted files with icat / Verify the recovered files

```

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 8 > recovered/Billing_Letter.doc

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 11 > recovered/confirmation.txt

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 15 > recovered/letter1.txt

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ icat evidence/Ch01InChap01.dd 17 > recovered/Regrets.doc

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ ls -lh recovered/
total 56K
-rw-rw-r-- 1 kali kali 24K Sep  3 18:23 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 227 Sep  3 18:24 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Sep  3 18:24 letter1.txt
-rw-rw-r-- 1 kali kali 23K Sep  3 18:24 Regrets.doc

(kali@kali)-[~/SBT-DF202_Lab3]
└─$ file recovered/*
recovered/Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title:
13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creatin
g Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri
Dec  9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
recovered/confirmation.txt: ASCII text, with CRLF line terminators
recovered/letter1.txt: ASCII text, with CRLF line terminators
recovered/Regrets.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title:
02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Crea
ting Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri
Dec  9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0

```

Examine the recovered text

```
(kali㉿kali)~[~/SBT-DF202_Lab3]
$ cat recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rrroi9

Thank you for your business

George

(kali㉿kali)~[~/SBT-DF202_Lab3]
$ cat recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
```

Block-level examination

[illegible]

The forensic image Ch01InChap01.dd was examined using The Sleuth Kit. The image was identified as a raw FAT12 filesystem with a size of 1,474,560 bytes and a 512-byte sector size. The **SHA-256 hash was recorded as 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122.**

Filesystem analysis identified four deleted/unallocated files: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. istat confirmed that their directory entries were marked Not Allocated while their associated sectors remained available.

The files were recovered successfully using `icat`. The recovered Word documents were identified as valid Microsoft Word Compound Document files, while the text files were identified as ASCII text. `blkcat` was also used to examine individual data sectors, confirming that sector-level content corresponded to the recovered deleted files.

The recovered text contained communications involving Laura, George, and Earl, including an FTP-related message and a separate meeting-related message. These findings demonstrate that deleted file contents can remain recoverable even after filesystem directory entries have been marked unallocated.

Part D

Binwalk

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ cd ~/SBT-DF202_Lab3

(kali㉿kali)-[~/SBT-DF202_Lab3]
$ binwalk J_ub_law.jpg | tee working/J_ub_law_binwalk.txt
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image directory: 8
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorporated"

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$
```

Embedded Content Analysis Using Binwalk

Binwalk was used to scan J_ub_law.jpg for embedded file signatures and additional data structures. The scan identified JPEG/EXIF data at offset 0, TIFF data at offset 12 within the EXIF structure, and an Adobe copyright string at offset 26,844. No separate embedded archive or additional recoverable file was identified.

Finding: The Binwalk results are consistent with a normal JPEG image containing EXIF/TIFF metadata and Adobe-related metadata. No additional embedded file requiring extraction was detected.

Part E Prepare and Examine the USB Image (120M.7z).

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ ls -lh 120M.7z
-rw-rw-r-- 1 kali kali 36M Sep  3 18:59 120M.7z
```

Compute Hashes

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ hashdeep -c md5,sha1 120M.7z
%%% HASHDEEP-1.0
%%% size,md5,sha1,filename
## Invoked from: /home/kali/SBT-DF202_Lab3/evidence
## $ hashdeep -c md5,sha1 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2810745018afaa2da31dc17a8eb590fca66eeef7,/home/kali/SBT-DF202_Lab3/evidence/120M.7z

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$
```

List the content of the zipfile

```
(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ 7z l 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

  Date       Time       Attr         Size   Compressed  Name
  ----       -
2021-09-22  22:59:31  D...A             0             0  120M
2021-09-22  22:59:31  ....A    124780544    36720256  120M/usb_fat_carving.001
2021-09-22  22:59:32  ....A         1685             0  120M/usb_fat_carving.001.txt
2021-09-22  22:59:32             124782229    36720256  2 files, 1 folders
```

inspect the accompanying .txt

```
(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ 7z x 120M.7z "120M/usb_fat_carving.001.txt" -so | tee ../working/usb_fat_carving_notes.txt
Created By AccessData® FTK® Imager 4.5.0.3

Case Information:
Acquired using: ADI4.5.0.3
Case Number: USB File Carving
Evidence Number: UB-002
Unique description: USB files used for file carving
Examiner: Frank Xu
Notes: University of Baltimore

Information for D:\cloud_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ_Team_Repository\DOJ_Case_Study\0_Basic_Computer_Skills_for_Forensics\file_carving\usb_image\120M\usb_fat_carving:

Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 15
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 243,712
[Physical Drive Information]
Drive Model: General UDisk USB Device
Drive Serial Number:
Drive Interface Type: USB
Removable drive: True
Source data size: 119 MB
Sector count: 243712
[Computed Hashes]
MD5 checksum: ba4a1d0ba49f4a6667b00a3b3e85e604
SHA1 checksum: bcc2d49fd49c9521ecb1739f6542c6bf327375ef

Image Information:
Acquisition started: Wed Sep 22 22:59:16 2021
Acquisition finished: Wed Sep 22 22:59:31 2021
Segment list:
D:\cloud_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ_Team_Repository\DOJ_Case_Study\0_Basic_Computer_Skills_for_Forensics\file_carving\usb_image\120M\usb_fat_carving.001

Image Verification Results:
Verification started: Wed Sep 22 22:59:32 2021
Verification finished: Wed Sep 22 22:59:32 2021
MD5 checksum: ba4a1d0ba49f4a6667b00a3b3e85e604 : verified
SHA1 checksum: bcc2d49fd49c9521ecb1739f6542c6bf327375ef : verified
```



```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ 7z e 120M.7z "120M/usb_fat_carving.001" -o.

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Size:          124780544
Compressed:    36720470

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ ls -lh usb_fat_carving.001
-rw-rw-r-- 1 kali kali 119M Sep 22  2021 usb_fat_carving.001
```

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ sha256sum usb_fat_carving.001 | tee ../hashes/usb_fat_carving.001.sha256
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9  usb_fat_carving.001

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ file usb_fat_carving.001
usb_fat_carving.001: DOS/MBR boot sector MS-MBR Windows 7 english at offset 0x163 "Invalid partition table" at offset 0x17b "Error loading operating system" at offset 0x19a "Missing operating system", disk signature 0xa1159a00; partition 1 : ID=0x0e, active, start-CHS (0x0,2,3), end-CHS (0x0,254,63), startsector 128, 243584 sectors

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ img_stat usb_fat_carving.001
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 124780544
Sector size: 512

(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ nmls usb_fat_carving.001
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot    Start      End      Length  Description
000:  Meta    0000000000  0000000000  0000000001  Primary Table (#0)
001:  _____ 0000000000  0000000127  0000000128  Unallocated
002:  000:000 0000000128  0000243711  0000243584  Win95 FAT16 (0x0e)
```

What we confirmed

SHA-256: 9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9

Image type: Raw forensic image

Sector size: 512 bytes

Total size: 124,780,544 bytes

Partition table: DOS/MBR

Partition: Win95 FAT16 (0x0e)

Partition start: Sector 128

Partition length: 243,584 sectors

Sectors 0–127 are unallocated before the partition.

The important point is that the FAT16 filesystem does not start at sector 0. It starts at sector 128, so we need to account for that offset when examining the filesystem.

Examine the the content of the usb

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ fsstat -o 128 usb_fat_carving.001 | tee ../working/usb_fat_fsstat.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT16
OEM Name: MSDOS5.0
Volume ID: 0x94105672
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): USB
File System Type Label: FAT16
Sectors before file system: 128
File System Layout (in sectors)
Total Range: 0 - 243583
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 241
* FAT 1: 242 - 479
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583
METADATA INFORMATION
-----
Range: 2 - 3889670
Root Directory: 2
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 60769
FAT CONTENTS (in sectors)
-----
512-515 (4) → EOF
516-519 (4) → EOF
520-523 (4) → EOF
70976-71211 (236) → EOF
71212-71215 (4) → EOF
```

Display Partitions

```
(kali㉿kali)-[~/SBT-DF202_Lab3/evidence]
$ fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                Boot Start    End Sectors  Size Id Type
usb_fat_carving.001p1 *      128 243711  243584 118.9M  e W95 FAT16 (LBA)
```

Find the delete files

```
(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ fls -o 128 -r usb_fat_carving.001 | tee ../working/usb_fat_fls.txt
r/r 3: USB (Volume Label Entry)
d/d 6: System Volume Information
+ r/r 519: WPSettings.dat
r/r * 7: _est
r/r 10: .dropbox.device
d/d * 13: old_File_Carving_files
+ r/r * 711: CarvedImage.jpg
+ r/r * 714: File_carving.docx
+ r/r * 717: File_Carving.pptx
+ r/r * 720: i-love-cats-cute-cats.jpg
r/r * 15: B_ub_poe4.bmp
r/r * 18: B_zoom-eubie-mono.bmp
r/r * 21: Ballardlab8.java
r/r * 24: brittLab10.java
r/r * 27: DO_example.doc
r/r * 30: DO_example2.doc
r/r * 33: G_BuiltForThis.gif
r/r * 35: G_zoom-sc.gif
r/r * 37: H_Form.html
r/r * 39: H_hello.html
r/r * 41: J_ub_law.jpg
r/r * 44: J_ub_night.jpg
r/r * 47: nps-2008-jean_outlook.pst
r/r * 50: P_CAS-zoom-6.png
r/r * 53: P_MSB_1_zoom.png
r/r * 57: pd_Evidence_search_techniques.pdf
r/r * 61: pd_Forensic_Report_Template.pdf
r/r * 64: pp_Number_Systems.pptx
r/r * 67: pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71: _tf_1.rtf
r/r * 74: T_Eubie-iphone-5-8.tiff
r/r * 78: T_youknowus-iphone-5-8.tiff
r/r * 81: W_axloadcomplete.wav
r/r * 84: W_speaker_test_sound.wav
r/r * 86: Z_file.zip
r/r * 88: 江莎莎行踪轨迹.doc
r/r * 91: nps-2008-jean_outlook.pst
v/v 3889667: $MBR
v/v 3889668: $FAT1
v/v 3889669: $FAT2
v/v 3889670: $OrphanFiles
```

Part F — Scalpel File Carving

Carving usb image

```
(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ sudo mousepad /etc/scalpel/scalpel.conf

(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ scalpel -c /etc/scalpel/scalpel.conf -o ../output/scalpel_carving usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.
ERROR: You have attempted to use a non-empty output directory. In order
to maintain forensic soundness, this is not allowed.
Aborting.

(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ rm -rf ../output/scalpel_carving
mkdir -p ../output/scalpel_carving

(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$ scalpel -c /etc/scalpel/scalpel.conf -o ../output/scalpel_carving usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/SBT-DF202_Lab3/evidence/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" → 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" → 9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 1 seconds.

(kali@kali)-[~/SBT-DF202_Lab3/evidence]
$
```

Show carved files

```
(kali㉿kali)-[~/SBT-DF202_Lab3]
$ tree output
output
├── scalpel_carving
│   ├── audit.txt
│   ├── jpg-0-0
│   │   ├── 00000000.jpg
│   │   ├── 00000001.jpg
│   │   ├── 00000002.jpg
│   │   ├── 00000003.jpg
│   │   ├── 00000004.jpg
│   │   ├── 00000005.jpg
│   │   ├── 00000006.jpg
│   │   └── 00000007.jpg
│   └── jpg-1-0
│       ├── 00000008.jpg
│       ├── 00000009.jpg
│       ├── 00000010.jpg
│       ├── 00000011.jpg
│       ├── 00000012.jpg
│       ├── 00000013.jpg
│       ├── 00000014.jpg
│       ├── 00000015.jpg
│       └── 00000016.jpg
4 directories, 18 files
```

Show audit log

```
(kali㉿kali)-[~/SBT-DF202_Lab3/output/scalpel_carving]
$ cat audit.txt

Scalpel version 1.60 audit file
Started at Fri Sep  4 06:48:11 2026
Command line:
scalpel -c /etc/scalpel/scalpel.conf -o ../output/scalpel_carving usb_fat_carving.001

Output directory: /home/kali/SBT-DF202_Lab3/output/scalpel_carving
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/kali/SBT-DF202_Lab3/evidence/usb_fat_carving.001"

The following files were carved:
File                Start                Chop                Length                Extracted From
00000010.jpg        3796992                NO                3148500                usb_fat_carving.001
00000009.jpg        425822                 NO                4875802                usb_fat_carving.001
00000008.jpg        335872                 NO                4965752                usb_fat_carving.001
00000002.jpg        6938624                NO                6868                  usb_fat_carving.001
00000001.jpg        5285888                NO                1659604                usb_fat_carving.001
00000000.jpg        3897344                NO                3048148                usb_fat_carving.001
00000004.jpg        15427584               NO                4223822                usb_fat_carving.001
00000003.jpg        12881920               NO                4256310                usb_fat_carving.001
00000006.jpg        19638272               NO                5229050                usb_fat_carving.001
00000005.jpg        17121280               NO                4248530                usb_fat_carving.001
00000007.jpg        21358592               NO                5183267                usb_fat_carving.001
00000016.jpg        36671488               NO                2889262                usb_fat_carving.001
00000015.jpg        36571136               NO                2989614                usb_fat_carving.001
00000014.jpg        36393610               NO                3167140                usb_fat_carving.001
00000013.jpg        36352448               NO                3208302                usb_fat_carving.001
00000012.jpg        35590996               NO                3969754                usb_fat_carving.001
00000011.jpg        35350242               NO                4210508                usb_fat_carving.001

Completed at Fri Sep  4 06:48:12 2026
```

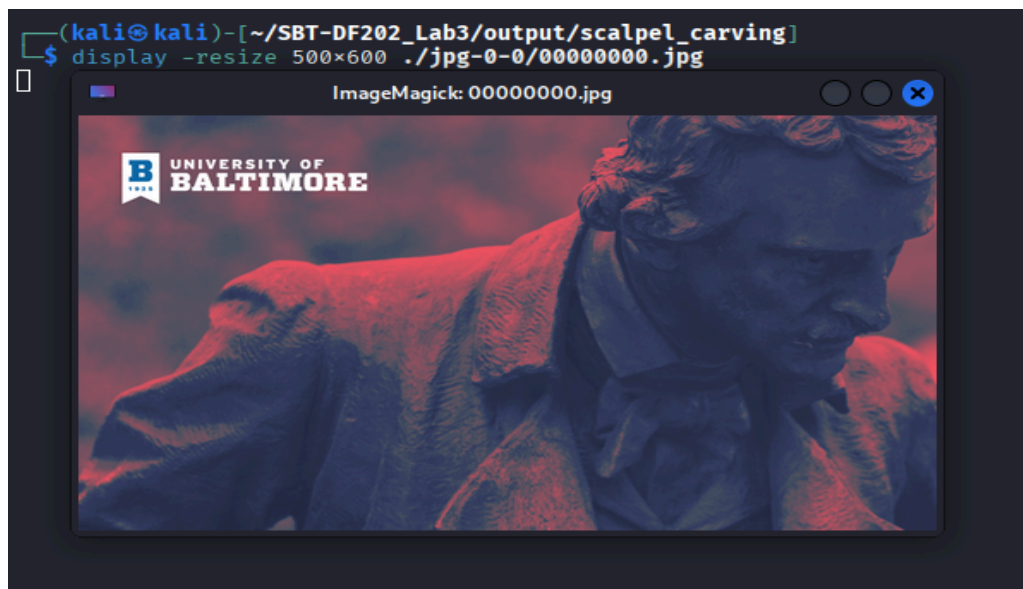
Scalpel Carving Results

The Scalpel audit log documented 17 JPEG files carved from usb_fat_carving.001. The carved files were assigned sequential names from 00000000.jpg through 00000016.jpg.

The recorded starting offsets ranged from 335,872 bytes to 36,671,488 bytes, with extracted lengths ranging from 6,868 bytes to 5,229,050 bytes. All 17 files were reported with Chop: NO.

The Scalpel audit file provides a repeatable record of the carving operation, including the source image, configuration file, output directory, file offsets, and extracted lengths.

Finding: JPEG data was successfully recovered directly from the raw USB image using file signatures, demonstrating recovery independent of normal filesystem metadata.



Scalpel successfully carved 17 JPEG files from the USB forensic image. The recovered images were stored in the jpg-0-0 and jpg-1-0 directories. Two carved JPEG files were opened for visual examination, demonstrating successful recovery of usable image content from the raw forensic image.

Analysis and Findings

The forensic analysis produced the following findings:

1. JPEG Hexadecimal Analysis

The JPEG sample J_ub_law.jpg was examined using xxd. The JPEG Start of Image (SOI) signature FF D8 and End of Image (EOI) signature FF D9 were identified. The hexadecimal data was successfully reconstructed into a JPEG file. The reconstructed file produced identical MD5 and SHA-256 hashes to the original, confirming bit-for-bit integrity.

2. JPEG Metadata Analysis

ExifTool identified the image as a JPEG captured using a NIKON D4 camera. Embedded metadata included timestamps, camera information, copyright information, keywords, and Adobe Photoshop processing history. These metadata elements provided useful investigative information but were treated cautiously because metadata can be altered.

3. Deleted-File Recovery

The FAT12 forensic image Ch01InChap01.dd was analyzed using The Sleuth Kit. Four deleted files were identified: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The files were marked as Not Allocated but their content remained recoverable using icat.

4. Sector-Level Analysis

blkcat was used to examine individual sectors associated with deleted files. Readable text was recovered from sectors containing confirmation.txt and letter1.txt, demonstrating that file content can remain present even after filesystem entries have been deleted.

5. Binwalk Analysis

Binwalk identified JPEG/EXIF data, TIFF data within the EXIF structure, and an Adobe copyright string. No separate embedded archive or additional recoverable file was identified.

6. USB Image and Scalpel Analysis

The USB forensic image was identified as a FAT16 filesystem. The Sleuth Kit identified multiple deleted files and directories. Scalpel was configured to recognize JPEG files using Exif and JFIF signatures.

Scalpel successfully carved 17 JPEG files from the raw USB image. The recovery was documented in audit.txt, including the starting offsets and extracted lengths of the carved files. Two recovered JPEG files were subsequently opened for visual examination.

Overall Finding

The investigation demonstrated that deleted or fragmented digital evidence may remain recoverable from raw storage media even when normal filesystem metadata is unavailable or marked as unallocated. The combination of filesystem analysis, hexadecimal examination, metadata analysis, sector-level recovery, and signature-based carving provided multiple methods for identifying and recovering forensic evidence.

Conclusion

This laboratory exercise successfully demonstrated the application of data carving and file recovery techniques in digital forensics. Hexadecimal analysis, metadata examination, filesystem analysis,

sector-level examination, embedded-content scanning, and signature-based file carving were used to identify and recover digital evidence.

The investigation successfully reconstructed and verified a JPEG file, recovered deleted files from a FAT12 forensic image, analyzed a FAT16 USB image, and used Scalpel to carve 17 JPEG files from raw storage data. Cryptographic hashing and audit logs were also used to support evidence integrity and reproducibility.

Overall, the exercise demonstrated that deleted files and useful digital evidence can remain recoverable from storage media even when filesystem metadata is missing or marked as unallocated. The results highlight the importance of combining multiple forensic techniques while preserving the integrity of the original evidence.